



Pudi Bhanu Swaroop

M.Tech Ophthalmic Engineering - TA

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Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Ophthalmic Engineering - TA	IIT Hyderabad	2026	8.59
B.Tech Information Technology	Sagi Ramakrishnam Raju Engineering College, Bhimavaram	2023	7.66
XII (Board of Intermediate Education, Andhra Pradesh)	Sasi Junior College, Tanuku	2019	9.94
X (Board of Secondary Education, Andhra Pradesh)	S.F.S. High School, Tanuku	2017	9.80

WORK EXPERIENCE

Graduate Researcher | BHIOVS Research Centre, LV Prasad Eye Institute (LVPEI), Hyderabad May 2025 – May 2026

- **Developed** IR imaging and optical simulation pipelines for **accommodation analysis in keratoconic eye datasets**.
- **Built image-processing workflows** using **Python, OpenCV, MATLAB** and **Zemax** for quantitative optical analysis.
- **Collaborated** with vision scientists and optometrists to **validate clinical workflows** and experimental protocols.

PROJECTS

Keratoconus Severity Classification System | [GitHub](#) | [Proof-Of-Concept](#)

- Developed Random Forest severity classifier achieving **93% accuracy** using 6-fold cross-validation on **clinical datasets**.
- Reduced **79 corneal variables to 3 diagnostic indicators** using Gini-based feature prioritization and preprocessing.
- Designed web-based analytics interface for **real-time severity prediction** and streamlined clinical evaluation workflows.

End-to-End RAG-Based AI Content Review Platform for Sunshine, IITH | [GitHub](#) | [Proof-of-Concept](#)

- Developed context-aware RAG assistant **reducing** verification time from **7 days to 30 minutes** with human-in-loop validation.
- **Optimized semantic retrieval** using Qdrant vector search, metadata indexing, and embedding-driven document pipelines.
- Designed **conversational AI interface** enabling faster contextual decisions across knowledge management workflows.

Zepto Delivery Delta Forecasting & Retention Optimization | [GitHub](#)

- **Built retention analytics pipeline** on **101K+ delivery records** to identify lateness as the primary churn driver.
- Developed Random Forest churn model achieving **0.79 AUC** and **0.83 F1-score** using SHAP-based explainability.
- Designed **CLV-based retention strategy** reducing intervention spend by **64%** against universal voucher campaigns.

PRODUCT & STRATEGY CASE STUDIES

Mitra.ai — AI Memory Assistant Case Study | [Pitch Deck](#)

- Led **primary research** across 50 respondents and 51 tools to identify **8 validated memory-management pain points**.
- **Designed AI memory workflow** with MCP connectors, contextual retrieval, and privacy-first multi-layer architecture.
- Built GTM and lifecycle strategy targeting campus-led adoption and bottom-up **enterprise expansion for retention**.

Tata Neu Gen-Z User's Engagement Strategy Case Study | [Pitch Deck](#)

- **Analyzed Gen Z behavioral trends** and identified habit-loop gaps driving Tata Neu's low repeat engagement.
- Designed **"Neu-Twin" smart lock-screen feature** enabling trend-led discovery, social rewards, and frictionless engagement.
- Defined **9 Hooked Model-based KPIs** tracking app opens, engagement, loyalty, and cross-brand interaction metrics.

SKILLS SUMMARY

- **Programming & Query Languages:** Python, SQL
- **AI/ML & Analytics:** Machine Learning, Exploratory Data Analysis (EDA), Statistical Analysis
- **Tools & Platforms:** MySQL, Jupyter Notebook, VS Code, Power BI
- **Product & Strategy:** Requirement Analysis, Product Metrics, User Research, Market Analysis, KPI Definition
- **Collaboration & Communication:** Presentation Skills, Cross-Functional Collaboration, Stakeholder Communication

POSITION OF RESPONSIBILITIES

PG Management Head | Sunshine, IIT Hyderabad

May 2025 - April 2026

- Increased Instagram reach by **20%** through structured collaboration across content, design, and publicity workflows.
- Led campus-wide engagement initiatives for **15+ wellness events** through high-visibility promotional campaigns.

PG Buddy | Sunshine, IIT Hyderabad

September 2024 - April 2025

- **Coordinated operations** for the National Well Being Conclave, enabling a seamless experience for **400+ delegates**.
- Contributed to **5+ campus wellness initiatives** by organizing workshops, expert talks, and student engagement activities.

ACHIEVEMENTS

- National Finalist (**Top 8 among 550+ teams**) in **Product Arena 3.0**, a national-level Product Management competition.
- Secured **Runner-Up** position in FCC Campus Case Challenge at IIT Hyderabad for **product strategy and problem-solving**.

EXTRA-CURRICULAR

Case Study Solving	Analyzed product, growth, and user-retention problems using behavioral frameworks and data-backed reasoning.
Cricket	Represented Vivekananda Hostel and IT Department in institute-level tournaments, strengthening teamwork and execution.
Reading	Explores books on psychology, product thinking, behavioral science, and personal development.